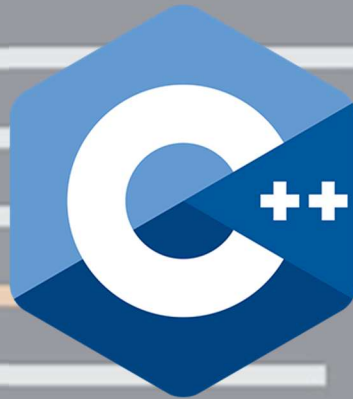




Now we C++

- By Aditya Varma

Preface

Welcome to Let's Learn C++

This book is written to help you learn C++ as a skill, not as a subject.

C++ is a language that gives you both control and structure. It allows you to work close to how programs actually run in memory, while also providing powerful tools to design large, organized systems. Because of this balance, learning C++ builds a deep understanding of programming that makes other languages easier to grasp.

The chapters in this book are arranged carefully. Each concept is introduced only when it is needed, and every idea is supported by simple programs that explain why something works, not just how to write it.

C++ can feel vast at first. It has many features, and it does not force you into one way of thinking. This book simplifies that journey by guiding you step by step, starting from the basics and gradually moving toward structured and object-oriented programming.

No prior programming knowledge is assumed. You are expected only to read, practice, and stay curious. This book is meant to be read slowly, like a guide. Type the programs. Modify them. Break them. Observe what changes.

Take your time. Ask questions. Make mistakes. That is how real understanding is built.

-Aditya Varma

License & Credits

License:

This work is licensed under the **Creative Commons Attribution–NonCommercial–ShareAlike 4.0 (CC BY-NC-SA 4.0)** license. You may share and adapt this material for non-commercial purposes, provided proper credit is given and derivative works use the same license.

Credits:

Certain icons, illustrations, and portions of reference content are adapted from publicly available educational platforms, including the Coding X mobile application. Proper credit is given where applicable. All original rights remain with their respective creators, and this material is used solely for educational purposes.

Index

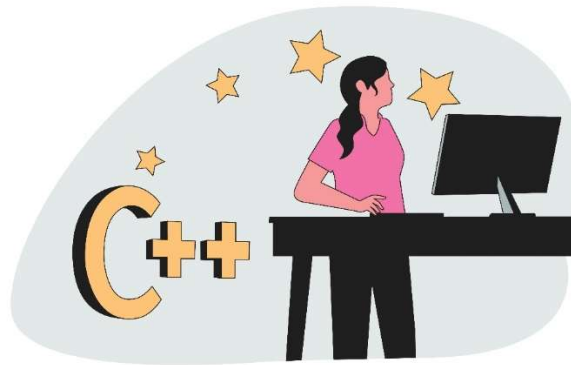
- 📑 Getting Started with C++
- 📑 First C++ Program & Boilerplate
- 📑 Escape Sequences & Comments
- 📑 The Language Building Blocks: Tokens
- 📑 Variables and Data Types in C++
- 📑 Operators in C++
- 📑 Input and Output in C++
- 📑 Program Flow in C++
- 📑 Conditionals Statements
- 📑 Looping Statements
- 📑 Loop Control Statements
- 📑 Arrays in C++
- 📑 Strings in C++
- 📑 Functions in C++
- 📑 OOP in C++
- 📑 Further Advance Topics

Getting Started with C++

C++ is a general-purpose programming language that sits at an interesting intersection: it gives you the power to work close to the machine while also letting you build large, structured, and maintainable software. It is often described as an extension of C, but over time it has grown into a language of its own, supporting multiple programming styles including procedural, object-oriented, and even generic programming.

At its core, C++ is about **control and efficiency**. When you write a C++ program, you are not just telling the computer *what* to do, but also influencing *how* it does it. You can manage memory directly, control performance, and decide how data is stored and accessed. This makes C++ both powerful and demanding. It does not hide complexity from you, but instead gives you the tools to understand and handle it.

Because of this, learning C++ is like learning how the engine of a machine works rather than just driving it. Once you understand it, other languages start to feel more intuitive.



History Of C++:

Like other programming languages popular today, C++ has quite a story behind it, stretching back to 1979. **Bjarne Stroustrup**, a Danish computer scientist working at Bell Labs, started work on a new programming language called '**C with Classes**' deriving from the original C language created by Dennis Ritchie

A large inspiration for the language was Simula, which is considered to be the first object-oriented programming language ever. While Stroustrup was programming for his Ph.D. thesis, he found Simula pleasant to program in, but the language was too slow for practical use. Therefore, he set out to add Simula-like features to C, thus making a language that was both performant and relatively high-level for the time. In this way, the 'C with Classes' programming language was born. Initially, C with Classes just added features to the C compiler.

In 1982, Stroustrup started to build the next version of his language. He built a new compiler based on the existing C compiler and added a ton of new features. After cycling through a couple of names, he ended up with 'C++'. The name comes from the C's incrementation operator, it means C plus one. In 1985, the first commercial implementation of C++ was released. The same year also saw the release of the first reference book of the language - The C++ Programming Language written by the creator himself

Stroustrup wanted the addition of object-oriented programming (OOP), which drove him to create C++. Therefore, the inclusion of OOP is the significant difference between C and C++. To summarize his interview with Big Think, the motivation for making C++ was increasing a program's maintainability.

To Conclude:

Although C++ is used in many industries and can be used to write almost anything, it particularly excels in delivering performance and using system resources efficiently. As we have said, it's an extension of the C programming language, therefore it retains a strong link to C, and will compile most C programs. Isn't that WONDERFUL?

Oh! That was quite a theory and a flashback into C++'s history. But, before you learn anything, you should know how it came into existence. From here on, we will move ahead to the more technical side and slowly start writing some code.

Difference Between C and C++:

To understand C++, it helps to see where it comes from. C is a procedural programming language, meaning it focuses on functions and step-by-step instructions. Programs in C are typically organized around functions that operate on data.

C++, on the other hand, expands this idea by introducing **object-oriented programming (OOP)**. Instead of only thinking in terms of functions, you can organize programs around **objects**, which combine data and behaviour together. This makes it easier to model real-world systems and manage large codebases.

Another key difference is that C++ provides additional features such as classes, inheritance, polymorphism, and function overloading. These features allow developers to write more modular and reusable code. While C gives you the raw tools, C++ adds layers that help you structure complex programs more effectively.

Despite these differences, C++ still retains most of C's capabilities. This means you can write low-level, efficient code just like in C, while also taking advantage of higher-level abstractions when needed. In a way, C++ is like a toolkit that includes both basic tools and advanced machinery, and it is up to the programmer to decide what to use.

Features of C++:

A lot of developers in different communities argue about the best programming language. Actually, it is impossible to identify which tool or technology is indeed the optimum variant for ALL problems since different languages are designed for different kinds of problems.

Nevertheless, the C++ language has some powerful arguments up its sleeve. In this section, we will look at some of the features of C++ and try to understand what makes it so popular and a top choice for various projects.

- **Performance:** Performance is directly related to speed. When it comes to speed, C++ has no match in today's list of popular languages. C++ is the Flash of the world of programming!

It is one of the fastest and most predictable languages out there, only contested by other low-level programming languages like Rust. The control provided by C++ over the system resources enables a skilled coder to write a program that is quicker and more powerful than a similar program written in another programming language.

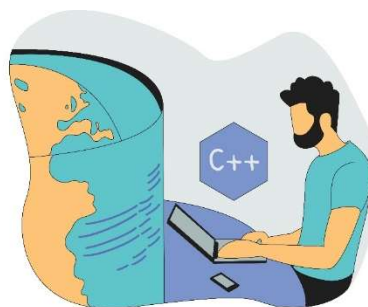
Interesting Fact: Most AAA game titles are written using C++ because these titles push the limits of existing hardware, so resource efficiency is very important.



- **Flexibility:** C++ is a multi-paradigm programming language. This means that it supports other styles such as procedural programming, in addition to object-oriented programming.

These paradigms are essentially different ways of looking at and solving a problem. Two different C++ programmers can solve the same problem in different ways. Paradigms can also be combined to get the most efficient result.

On the other side, having different ways of solving problems makes C++ more complex, but at the same time, more powerful as well.



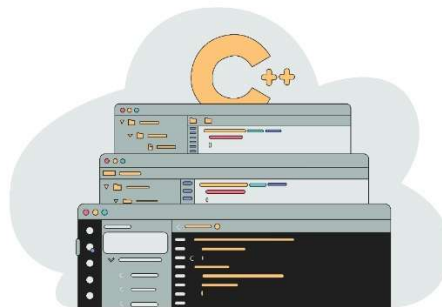
- **Syntax:** When it comes to syntax, it is a bit harder to choose the best option. Today, the majority of popular programming languages have pretty nice syntax. Unlike some other languages, C++ has quite an intuitive syntax, which is actually not so hard to comprehend, by looking at the code a bit more creatively.

Hence, the prize named "Easiest Syntax" doesn't go to C++, since it has more elements to write in contrast to some other languages. Fortunately, even with more code and intuitive syntax, C++ code is more structured and follows the same set of rules. Therefore, despite its length, C++ has a pretty straightforward syntax.



- **Large Ecosystem:** The language has been around for almost 40 years, which means that most of the software problems have already been solved by open-source libraries and frameworks. C++ has a ton of developers that use it, upgrade it, and write open-source libraries.

Therefore, in simple words, while learning or using the language, you can draw on the work done by these people already.



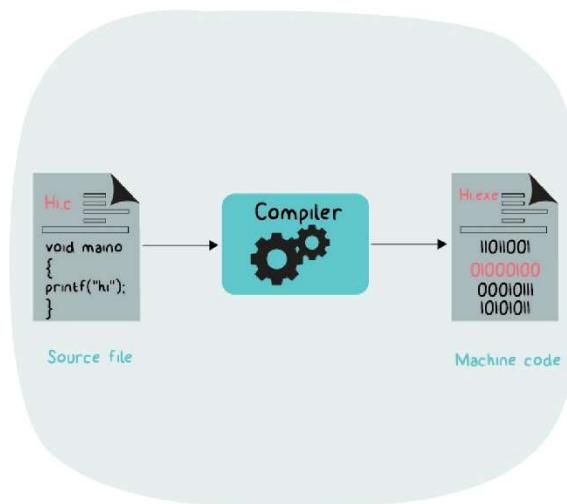
C++ is an exceptional language when it comes to its abilities. It lets us do virtually anything possible and impossible. It is a technology that can be used to accomplish a tremendous variety of tasks.

A large number of jobs available is a reflection of the long-standing popularity of C++. Not only do many new applications get coded in C++, but the huge range of programs already built during the language necessitate hiring coders to keep them current and updated. Hence, C++ developers are welcome to nearly every software development company.

How C++ Works?

When you write a C++ program, you do not directly communicate with the computer's hardware. Instead, you write code in a human-readable form called **source code**, usually saved in files with a `.cpp` extension. This code must be translated into a form that the computer can understand and execute.

This translation happens through a process called **compilation**.



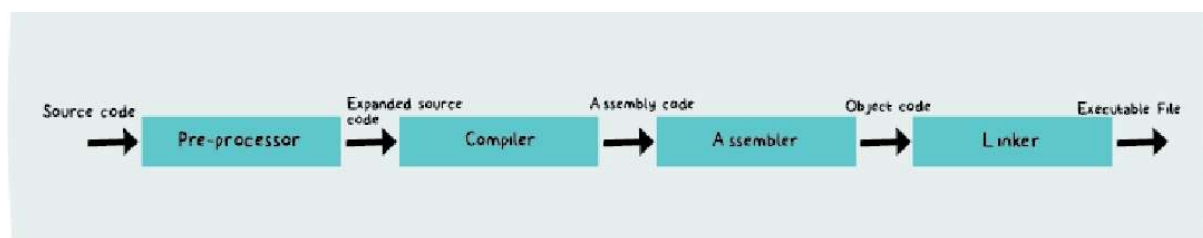
Compiled languages are converted directly into machine code that the processor can execute. As a result, they tend to be faster and more efficient to execute than interpreted languages.

Steps in Compilation:

The first step involves the compiler taking your source code and checking it for errors. If the code is valid, it is converted into an intermediate form called **object code**. This object code is not yet a complete program, but it contains machine-level instructions.

In the next step, a **linker** combines this object code with other necessary pieces, such as libraries, to produce the final **executable file**. This executable file is what you actually run on your system.

So, the journey of a program can be thought of as a transformation:



Where is C++ used?

Remember, C++ is a general-purpose programming language? This means it is used in most places! And that's the reason why C++ has held its grip even after 35+ years of its release. C++ can be used for; Operating Systems Development, Game Development, Computer Graphics, Cloud/Distributed Systems, Compiler Development, And more... The list goes on! Let's dive deep.

- **Operating Systems:** Be it Microsoft Windows or Mac OSX or Linux - all of them are programmed in C++. No doubt, right?

Speed and security are the main factors for an OS, and C++ is the master of those features, C/C++ is the backbone of all well-known operating systems owing to the fact that it is a strongly typed and fast programming language which makes it an ideal choice for developing an operating system.

- **Game Development:** Game Development is one of the fields that is said to have one of the brightest futures in the upcoming decade. Well, believe it or not, C++ is a direct ticket to a billion-dollar game development industry.

To tackle big games in the larger gaming companies, knowing C++ is critical. It's fast, the compilers and optimizers are solid, and you get a lot of control over memory management. It has extensive libraries, which come in handy for designing and powering complex graphics.

- **Computer Graphics:** All graphics applications require fast rendering and just like the case of web browsers, C++ helps in reducing the latency. Software that employs computer vision, digital image processing, high-end graphical processing - they all use C++ as the backend programming language.

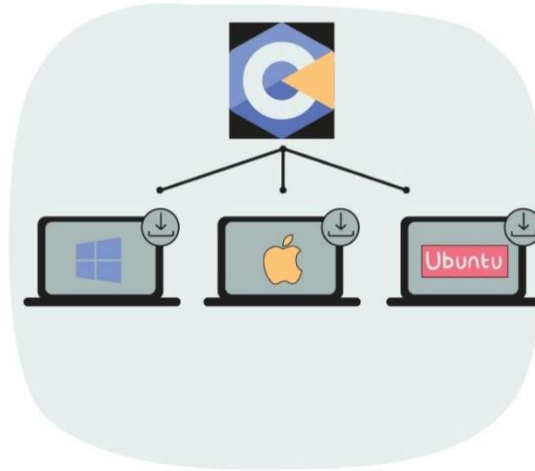
Even the popular games that are heavy on graphics use C++ as the primary programming language. The speed that C++ offers in such situations helps the developers in expanding the target audience because an optimized application can run even on low-end devices that do not have high computation power available.

- **Cloud/Distributed Systems:** Large organizations that develop cloud storage systems and other distributed systems also use C++ because it connects very well with the hardware and is compatible with a lot of machines. Cloud storage systems use scalable file systems that work close to the hardware.
- **Compiler Development:** A compiler is responsible for not only converting the source code to machine code but also performs various checks to ensure the code is correct and error-free. The compilers of various programming languages use C and C++ as the main programming language.



Installing C++ and its IDE:

To work with C++ on your machine, first, you will need to install the C++ Compiler in your system. It is unlikely that your system will be shipped with C++ Compiler.



Almost every C++ programmer uses the simplest way for getting this by using the GCC compiler. But first, would you like to know what GCC is? GCC stands for GNU Compiler Collection, it is an open-source compiler that supports the execution of multiple programming languages like C, C++, Fortran, Golang, and more.

GCC has been ported to multiple platforms and hence it helps in building C++ programs on a wide variety of systems. Currently, GCC's appropriate version for multiple platforms is available widely and for free.

But we are going to use something more efficient i.e., “Codeblocks”.

Installing Code::Blocks with MinGW:

Before we start writing C++ programs, we need a place to write and run them. For this, we will use Code::Blocks, which comes with a built-in C/C++ compiler called MinGW. Follow the steps carefully.

Step 1: Download Code::Blocks

- Open your browser
- Search for: Code::Blocks official website
- Open the website:
👉 Code::Blocks
- Go to the Downloads section
- Click on “Download the binary release”

Step 2: Choose the Correct Version

You will see multiple options. Choose the one that includes MinGW.

- Look for a file named similar to:
codeblocks-20.03mingw-setup.exe

⚠ Important:

- Always choose the version with “mingw” in the name
- This version already contains the C compiler
- No need to install anything separately

Step 3: Install Code::Blocks

- Double-click the downloaded .exe file
- Click Next → I Agree → Next
- Keep default settings
- Click Install

Once installation completes:

- Click Finish

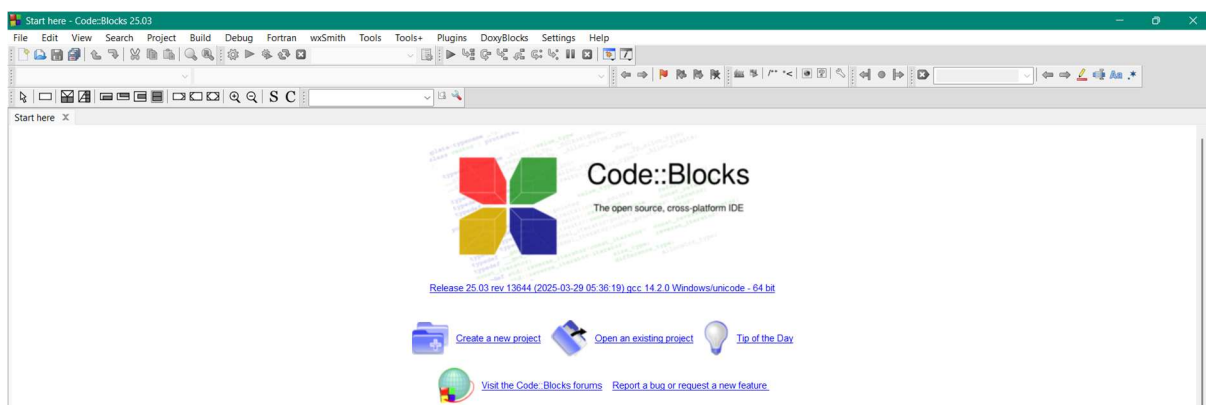
Step 4: First Launch Setup

- Open Code::Blocks
- It may ask to select a compiler

If prompted:

- Choose: GNU GCC Compiler
- Click Set as default
- Click OK

The setup is done and screen will look like this:



Creating files in Code::Blocks:

Step 1: Create Your First Project

- Click on Create a new project
- Select Console Application
- Click Go → Next
- Choose language: C++
- Click Next

Step 2: Set Project Details

- Enter a Project Title (e.g., MyFirstProgram)
- Choose a folder location
- Click Next → Finish

Step 3: Write Your First Program

You will see a default file (main.cpp).

- Inside it, you can write your first program
- Or modify the existing code

Step 4: Build and Run the Program

- Click on Build and Run (green play button)
OR press F9
- Output will appear in a console window

At this stage, do not worry about:

- How the compiler works internally
- Complex settings
- Advanced configurations

Right now, your goal is simple:

Write → Run → Observe

First C++ Program & Boilerplate

Let's begin with the classic first step into any programming language. It may look small, but it opens the door to everything that follows.

```
1 #include <iostream>
2
3 int main() {
4     std::cout << "Hello, World!";
5     return 0;
6 }
```

When this program runs, it simply prints:

Hello, World!

At first glance, this might feel like a few strange symbols stitched together. But each line has a purpose, and together they form the basic skeleton of almost every C++ program you will write.

Understanding the Structure of a C++ Program:

A C++ program is not just random lines of code. It follows a structure, almost like a recipe. Each part has a role, and removing or misplacing one piece can break the whole program.

Let's gently unwrap each component.

1. `#include <iostream>`: Bringing in Tools

This line is a **preprocessor directive**. It tells the compiler to include a file before the program is compiled. The file `iostream` stands for **input-output stream**. It contains definitions for objects that allow your program to take input and display output.

Think of it like opening a toolbox before starting work. Without it, your program would not know how to print anything on the screen.

2. What is `iostream`?

`iostream` is a standard library file in C++ that provides functionality for input and output operations.

It gives you access to:

- `cout` → for output (printing to screen)
- `cin` → for input (taking data from user)

Without including `iostream`, using `cout` or `cin` would result in an error because the compiler would not recognize them.

3. What is std?

In C++, many standard features are grouped under something called a **namespace**. The standard library belongs to the namespace called `std` (short for *standard*).

So, when you write: `std::cout` you are telling the compiler: “Use `cout` from the standard namespace.”

This avoids confusion in large programs where different libraries might have functions or objects with the same name.

4. `int main()`: The Starting Point

Every C++ program begins execution from the `main()` function.

```
int main() {  
    // code  
}
```

- `main` is the entry point of the program
- `int` means the function returns an integer value
- The `{}` braces contain the body of the program

No matter how big your program becomes, execution always starts from here.

5. `cout`: Displaying Output

Inside the `main()` function, we wrote: `std::cout << "Hello, World!";`

Here:

- `cout` stands for **console output**
- `<<` is the insertion operator (it sends data to output)
- `"Hello, World!"` is the text to display

So this line simply tells the computer: “Print this message on the screen.”

6. `return 0`; Ending the Program

At the end of `main()`, we often write: `return 0;`

This indicates that the program has executed successfully.

- `0` typically means success
- Any non-zero value can indicate an error

While modern compilers may allow you to omit it, it is good practice to include it for clarity.

Making It Simpler: Using namespace `std`

Writing `std::cout` every time can feel repetitive, especially for beginners. C++ allows you to simplify this using: `using namespace std;`

This tells the compiler: “Use the standard namespace by default.”

Now you can write a cleaner version of the same program.

C++ Boilerplate (Basic Template):

```
1  #include <iostream>
2
3  int main() {
4      // Program codes goes here
5      return 0;
6  }
```

This is the **basic boilerplate** of a C++ program.

Final Thought:

If a C++ program were a stage play:

- `#include` opens the backstage doors
- `iostream` brings in the actors for input/output
- `std` tells you which troupe they belong to
- `main()` is where the play begins
- `cout` is the actor speaking to the audience

Right now, it's just one line of dialogue. Soon, you'll be directing entire performances.

Escape Sequences & Comments

// Comments in C++:

When you write programs, you are not just communicating with the computer, but also with yourself and others who may read your code later. This is where **comments** come in. Comments are lines in a program that are ignored by the compiler. They do not affect how the program runs, but they make the code easier to understand.

In real-world programming, code is rarely written once and forgotten. It is read, modified, debugged, and improved over time. Comments act like small notes left in the code, explaining *why something is done*, not just *what is written*. A well-commented program feels guided, while a program without comments can feel like walking through a maze without signs.

C++ provides two types of comments:

- **Single-line comments**, which start with `//` and continue till the end of the line
- **Multi-line comments**, which start with `/*` and end with `*/`, allowing you to write comments across multiple lines

Comments are especially useful for:

- Explaining logic
- Temporarily disabling code during testing
- Making programs readable for others

Lesson Program: Comments in C++

Source Code: Comments.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // Printing a welcome message
7      cout << "Welcome to C++ Programming!\n";
8
9      // Printing another line
10     cout << "This program demonstrates comments usage.\n";
11
12     /*
13      Multi-line comment:
14      The following lines perform simple addition
15      and display the result
16     */
17
18     int a = 10;    // First number
19     int b = 20;    // Second number
20     int sum = a + b; // Adding numbers
21
22     // Displaying the result
23     cout << "Sum = " << sum << endl;
24
25     return 0;    // Program ends successfully
26 }
```

// Escape Sequence:

When printing text, sometimes we need more control than just plain words. We may want to move to a new line, insert a tab space, print quotation marks, or format the output neatly. This is where **escape sequences** come into play.

An escape sequence is a combination of characters that begins with a backslash \. It tells the compiler to perform a special action instead of printing the characters as they are. Think of escape sequences as hidden instructions embedded inside strings. They allow you to control how the output appears on the screen.

Why Do We Need Escape Sequences?

If you write: `cout << "Hello World";` everything appears on one line.

But what if you want:

```
Hello
World
```

You cannot just press Enter inside the string. Instead, you use:

```
cout << "Hello\nWorld";
```

Here, `\n` tells the program to move to a new line.

Where and How Are They Used?

Escape sequences are mainly used inside **string literals** (text inside quotes "). They modify how the output is displayed.

For example:

- Formatting output
- Printing special characters
- Structuring readable console output

List of Escape Sequences in C

Here are the most useful escape sequences:

- `\n` → New line
- `\t` → Horizontal tab (space gap)
- `\b` → Backspace (removes previous character)
- `\r` → Carriage return (moves cursor to beginning of line)
- `\\` → Prints a backslash \
- `\"` → Prints double quotes "
- `\'` → Prints single quote '
- `\0` → Null character (end of string)

Lesson Program: Escape Sequence in C++**Source Code:** EscapeSequence.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // Demonstrating Escape Sequences in C++
7
8      cout << "Demonstrating Escape Sequences in C++\n\n";
9
10     // 1. New Line
11     cout << "1. New Line (\\n):\n";
12     cout << "Hello\nWorld\n\n";
13
14     // 2. Tab Space
15     cout << "2. Tab Space (\\t):\n";
16     cout << "Hello\tWorld\n\n";
17
18     // 3. Backspace
19     cout << "3. Backspace (\\b):\n";
20     cout << "Hello\b World\n\n";
21
22     // 4. Carriage Return
23     cout << "4. Carriage Return (\\r):\n";
24     cout << "Hello World\rHi\n\n";
25
26     // 5. Backslash
27     cout << "5. Backslash (\\\\):\n";
28     cout << "\\n\n";
29
30     // 6. Single Quote
31     cout << "6. Single Quote (\\'):\n";
32     cout << "'\n\n";
33
34     // 7. Double Quote
35     cout << "7. Double Quote (\\\"):\n";
36     cout << "\"C++\"\n\n";
37
38     // 8. Alert (may produce sound)
39     cout << "8. Alert (\\a):\n";
40     cout << "\a\n";
41
42     return 0;    // Program ends
43 }
```

The Language Building Blocks: Tokens

Before a C++ program can be understood by the compiler, it must be broken down into smaller pieces. These smallest individual units of a program are called **tokens**. Just like a sentence is made up of words and punctuation, a C++ program is made up of tokens. The compiler reads your code not as full lines or paragraphs, but as a sequence of these meaningful building blocks.

Whenever you write a program, whether it is a simple `cout` statement or a complex function, everything you type is ultimately divided into tokens. This is why understanding tokens helps you understand how the compiler “sees” your code.

In simple terms, **a token is the smallest unit in a C++ program that has meaning to the compiler.**

Why Are Tokens Important?

Tokens are important because they form the basic structure of any program. If tokens are used incorrectly, the compiler will not understand the code and will generate errors. For example, writing a variable name incorrectly or misusing an operator can break the entire program.

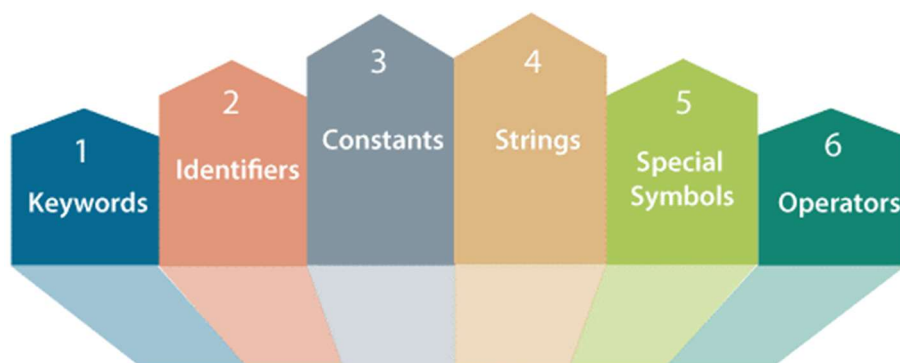
By learning tokens, you begin to see programming not as random syntax, but as a well-organized system where every symbol has a purpose. It also helps in debugging, because you can identify exactly which part of the code is causing an issue.

Types of Tokens in C++:

C++ mainly has **six types of tokens**, each playing a specific role:

- **Keywords:** Reserved words with special meaning (e.g., `int`, `return`, `if`)
- **Identifiers:** Names given to variables, functions, etc. (e.g., `sum`, `main`)
- **Constants (Literals):** Fixed values (e.g., `10`, `3.14`, `'A'`)
- **Strings:** Sequence of characters inside quotes (e.g., `"Hello"`)
- **Operators:** Symbols that perform operations (e.g., `+`, `-`, `*`, `=`)
- **Punctuators (Special Symbols):** Symbols used for structure (e.g., `;`, `{}`, `()`)

Together, these tokens form every valid C++ program.



Character Set of C++

To form tokens, C++ uses a defined set of characters known as the **character set**. These are the basic symbols that you can use to write any program.

The C++ character set includes:

- Uppercase: A to Z
- Lowercase: a to z
- Digits: 0 to 9
- Special characters: ! " # \$ % & ' () * + - . : ; ` ~ = < > { } [] ^ _ \ /
- Other special characters: Blank space, tab, etc. (Compiler ignores white spaces unless they are part of string)

These characters combine to form tokens, and tokens combine to form complete programs.

// Keywords & Identifiers:

In every C++ program, the words you write are not all treated the same. Some words are already defined by the language itself, while others are created by the programmer. This brings us to two important types of tokens: **keywords** and **identifiers**.

C++ Keywords:

Keywords are **reserved words** that have a fixed meaning in C++. They are predefined by the language and cannot be used for any other purpose, such as naming variables or functions.

For example, words like `int`, `float`, `return`, and `if` are keywords. Each of these has a specific role. When you write `int`, the compiler immediately understands that you are declaring an integer variable. These meanings are built into the language and cannot be changed.

C++ has a larger set of keywords compared to C because it supports advanced features like object-oriented programming. All keywords are written in lowercase and must be used exactly as defined.

Keywords act like the grammar of the language. They give structure to your program and tell the compiler how to interpret different parts of the code.

List of Keywords:

<code>asm</code>	<code>auto</code>	<code>bool</code>	<code>break</code>	<code>case</code>	<code>catch</code>
<code>char</code>	<code>class</code>	<code>const</code>	<code>const_cast</code>	<code>condition</code>	<code>default</code>
<code>delete</code>	<code>do</code>	<code>double</code>	<code>dynamic_cast</code>	<code>else</code>	<code>enum</code>
<code>explicit</code>	<code>export</code>	<code>extern</code>	<code>false</code>	<code>float</code>	<code>for</code>
<code>friend</code>	<code>goto</code>	<code>if</code>	<code>inline</code>	<code>int</code>	<code>long</code>
<code>mutable</code>	<code>namespace</code>	<code>new</code>	<code>operator</code>	<code>private</code>	<code>protected</code>
<code>public</code>	<code>register</code>	<code>reinterpret_cast</code>	<code>return</code>	<code>short</code>	<code>signed</code>
<code>sizeof</code>	<code>static</code>	<code>static_cast</code>	<code>struct</code>	<code>switch</code>	<code>template</code>
<code>this</code>	<code>throw</code>	<code>true</code>	<code>try</code>	<code>typedef</code>	<code>typeid</code>
<code>typename</code>	<code>union</code>	<code>unsigned</code>	<code>using</code>	<code>virtual</code>	<code>void</code>
<code>volatile</code>	<code>wchar_t</code>	<code>while</code>			

C++ Identifiers:

Identifiers are the names given to elements created by the programmer. These include variables, functions, arrays, classes, and more.

For example: `int age; float height;`

Here, age and height are identifiers. They are chosen by the programmer to represent data in a meaningful way. Unlike keywords, identifiers do not have predefined meanings. Their purpose depends entirely on how they are used in the program. Good identifiers make code more readable and easier to understand.

Rules for Defining Identifiers in C++:

While identifiers give you flexibility, there are certain rules that must be followed:

1. An identifier must begin with a letter (A–Z or a–z) or an underscore (_)
2. It can contain letters, digits (0–9), and underscores
3. It cannot start with a digit
4. It cannot be a keyword
5. Identifiers are case-sensitive (age and Age are different)
6. No special symbols like @, #, %, or spaces are allowed

✓ Example Valid Identifiers:

- `sum, totalMarks, _count, Age1`

✗ Example Invalid Identifiers:

- `int` (keyword), `2ndValue` (starts with digit), `my-name` (special character)

Lesson Program: Keywords and Identifiers in C++

Source Code: KeywordsAndIdentifiers.cpp

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // int, float, char are keywords
7      // age, height, grade are identifiers
8
9      int age = 23;           // 'int' is a keyword, 'age' is an identifier
10     float height = 5.9;     // 'float' is a keyword, 'height' is an identifier
11     char grade = 'A';       // 'char' is a keyword, 'grade' is an identifier
12
13     // cout and endl are part of standard library (used for output)
14
15     cout << "Age: " << age << endl;
16     cout << "Height: " << height << endl;
17     cout << "Grade: " << grade << endl;
18
19     // return is a keyword
20     return 0;
21 }
```

// Constants and Variables in C++

Every C++ program works with **data**. Data is the information that a program stores, processes, and displays. In C++, this data is mainly classified into two types: **constants** and **variables**.

Constants in C++:

A constant is a **fixed value** that does not change during the execution of a program. Once defined, its value remains the same throughout. Constants are used when certain values are meant to stay unchanged, such as mathematical values or fixed quantities.

Constants can be of different types:

- **Integer constants** → 10, -25, 0
- **Floating-point constants** → 3.14, -0.001, 2.5e3
- **Character constants** → 'A', 'z', '1'
- **String constants** → "Hello", "C++ Programming"

In C++, constants can also be defined using the `const` keyword.

Example:

```
const int days = 7;
```

Here, `days` is a constant whose value cannot be changed.

Variables in C++:

A variable is a **named location in memory** used to store data that can change during program execution. Unlike constants, variables can be modified as the program runs. Before using a variable, it must be declared with a data type, which tells the compiler what kind of data it will store.

Example:

```
int age = 20;  
float height = 5.9;  
char grade = 'A';
```

Here:

- `age`, `height`, and `grade` are variables
- Their values can change during execution

Variables are essential because they allow programs to store and manipulate dynamic data.

Rules for Variables

The rules for naming variables in C++ are the same as identifiers:

- Must begin with a letter or underscore
- Can contain letters, digits, and underscores
- Cannot be a keyword
- Case-sensitive
- No special symbols or spaces allowed

Lesson Program: Constants and Variables in C++**Source Code:** ConstantsAndVariables.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // Constant (value cannot be changed)
7      const float PI = 3.14;
8
9      // Variables (values can change)
10     int days = 7;
11     int age = 23;
12     float height = 5.9;
13     char grade = 'A';
14
15     // Displaying constant value
16     cout << "PI = " << PI << endl;
17
18     // Displaying variable values
19     cout << "Days in a week = " << days << endl;
20     cout << "Age = " << age << endl;
21     cout << "Height = " << height << endl;
22     cout << "Grade = " << grade << endl;
23
24     return 0;
25 }
```

// Constants, Strings, Operators, and Punctuators in C++:

In addition to keywords and identifiers, C++ programs are built using other important tokens such as **constants**, **strings**, and **punctuators**. These tokens help represent values, text, and the structure of a program. Each of them plays a simple but essential role in writing meaningful code.

Constants in C++:

Constants are **fixed values** that do not change during the execution of a program. Once a constant is defined, its value remains the same throughout.

For example: 10, 3.14, 'A'

These values are directly used in programs without needing a variable. Constants can be of different types:

- **Integer constants** → 10, 100
- **Floating-point constants** → 3.14, 2.5
- **Character constants** → 'A', 'b'

Constants make programs simple when values do not need to change. They are used to represent fixed data such as marks, grades, or predefined numbers.

Strings in C++:

Strings are a sequence of characters enclosed in **double quotes**.

For example: "Hello", "C++ Programming"

Strings are mainly used to display messages or text in a program. They are commonly used with output statements like `cout`. Unlike character constants (which use single quotes), strings always use double quotes and can contain multiple characters.

Operators in C++:

Operators are **symbols used to perform operations** on data. They tell the compiler to carry out specific tasks such as addition, comparison, or assignment.

For example: +, -, *, /, =

Operators are used to:

- Perform calculations
- Compare values
- Assign values

They are essential for writing logic in programs.

Punctuators in C++:

Punctuators are **special symbols** used to organize and structure a C++ program. They help the compiler understand how different parts of the code are separated and grouped.

Some common punctuators include: `;` `()` `{}` `,` `[]` `.`

For example:

- `;` marks the end of a statement
- `{}` define a block of code
- `()` are used in functions

Without punctuators, the structure of a program would be unclear, and the compiler would not be able to interpret it correctly.

Lesson Program: Other tokens in C++

Source Code: OtherTokens.cpp

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // Constants
7      int a = 10;           // 10 is an integer constant
8      int b = 5;           // 5 is an integer constant
9      char ch = 'A';       // 'A' is a character constant
10
11     // String
12     cout << "Welcome to C++ Programming\n";
13
14     // Operators
15     int sum = a + b;       // '+' is an operator
16     int diff = a - b;     // '-' is an operator
17
18     // Displaying values
19     cout << "Sum = " << sum << endl;
20     cout << "Difference = " << diff << endl;
21     cout << "Character = " << ch << endl;
22
23     // Punctuators used:
24     // ; -> end of statement
25     // {} -> block of code
26     // () -> function
27
28     return 0;
29 }
```

Variables and Data Types in C++

Until now, programs were mainly printing messages. But real programs do much more than that. They **store values, use them, and sometimes change them over time**.

Think about simple real-life situations:

- A student has marks
- A shop has prices
- A program calculates a result

In all these cases, some values need to be stored and used later. A computer does not “remember” things like we do. Instead, it stores everything in its memory. This is where **variables** come into the picture.

What is a Variable?

A variable is a **named location in memory** where a value is stored. When a program runs, the computer uses its memory to store data. Instead of remembering memory locations (which are complex), we give them simple names. These names are called **variables**.

In simple terms: A variable is a **container that stores data**

Understanding Variables with Memory:

You can imagine memory like a collection of boxes  :

- Each box has a **name** → variable
- Each box stores a **value**
- Each box has a **type** → defines what kind of data it can hold

For example: `int age = 21;`

Here:

- `int` → type (integer)
- `age` → variable name
- `21` → value stored

Declaring a Variable:

Declaration means telling the compiler: “I want to create a variable of this type with this name”

```
int age;  
float salary;  
char grade;
```

At this stage:

- Memory is reserved
- No value is stored yet

Assigning a Value:

Assignment means giving a value to a variable.

```
age = 21;  
salary = 55000.0;  
grade = 'A';
```

This stores the values in memory.

Initializing a Variable:

Initialization means declaring and assigning a value at the same time.

```
int year = 2025;  
float pi = 3.14;  
char section = 'B';
```

This is the most common and recommended way.

Where Are Variables Used?

Variables are used:

- To store input values
- To perform calculations
- To hold results
- To control program flow

Almost every program depends on variables.

Rules for Naming Variables:

Variable names follow the same rules as identifiers:

- Must start with a letter (a–z, A–Z) or underscore (_)
- Can contain letters, digits, and underscores
- Cannot start with a digit
- Cannot be a keyword
- No spaces or special symbols
- Case-sensitive (age and Age are different)

Making a Variable Constant

Sometimes, we do not want a value to change. In such cases, we use **constants**. C++ provides **two ways** to create constants:

1. Using const Keyword:

```
const float PI = 3.14;
```

- Value cannot be changed later
- Type-safe (follows data type rules)

2. Using #define (Preprocessor Constant):

#define DAYS 7

- Defined before compilation
- No data type is specified
- Replaces value everywhere in code

Difference (Basic Idea):

- `const` → behaves like a variable (recommended in C++)
- `#define` → simple text replacement

Lesson Program: Variables in C++

Source Code: Variables.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  // Preprocessor constant
5  #define DAYS 7
6
7  int main() {
8
9      // Declaration
10     int age;
11     float salary;
12     char grade;
13
14     // Assignment
15     age = 21;
16     salary = 55000.0;
17     grade = 'A';
18
19     // Initialization
20     int year = 2025;
21     float pi = 3.14;
22     char section = 'B';
23
24     // Constant using const
25     const float PI = 3.14159;
26
27     // Using variables
28     cout << "Age: " << age << endl;
29     cout << "Salary: " << salary << endl;
30     cout << "Grade: " << grade << endl;
31
32     cout << "Year: " << year << endl;
33     cout << "Pi: " << pi << endl;
34     cout << "Section: " << section << endl;
35
36     // Using constants
37     cout << "Days in a week (#define): " << DAYS << endl;
38     cout << "Value of PI (const): " << PI << endl;
39
40     return 0;
41 }
```

// Data Types in C++

In C++, every variable stores some kind of data. But not all data is the same. Some values are whole numbers, some are decimal numbers, and some are characters or text. To handle this properly, C++ uses **data types**.

A **data type** defines:

- What kind of data a variable can store
- How much memory it will use
- What range of values it can hold

In simple terms: A data type tells the computer **what type of value is being stored**

Why Do We Need Data Types?

Computers manage memory very carefully. If you store a small number, it should not take too much space. If you store a large number or decimal value, more memory may be needed.

Data types help:

- Use memory efficiently
- Perform correct operations
- Avoid errors

For example, storing 10 is different from storing 10.5, and both are different from storing 'A'.

Classification of Data Types in C++

1. Primary (Fundamental) Data Types in C++:

Primary data types are the basic built-in types supported directly by the compiler.

They include:

- `int` → stores integers (e.g., 10, -5)
- `float` → stores decimal values (e.g., 3.14)
- `double` → stores decimal values with higher precision
- `char` → stores a single character (e.g., 'A')
- `bool` → stores logical values (`true` or `false`)

2. Derived Data Types:

Derived data types are formed using primary data types.

Examples:

- Arrays
- Pointers
- Functions

They help in handling collections of data and memory.

3. User-defined Data Types:

User-defined data types are created by the programmer.

Examples:

- struct
- union
- enum
- typedef

They help in organizing complex data.

Note on Size and Range

Each data type uses a specific amount of memory and has a defined range. This may vary depending on the system and compiler, but the purpose remains the same: Efficient storage and proper data handling

Lesson Program: Data Types in C++

Source Code: DataTypes.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // Primary data types
7
8      int age = 23;           // Integer type
9      float pi = 3.14;       // Float type
10     double big = 123456.789; // Double type
11     char grade = 'A';       // Character type
12     bool isStudent = true;  // Boolean type
13
14     // Displaying values
15     cout << "Integer (age): " << age << endl;
16     cout << "Float (pi): " << pi << endl;
17     cout << "Double (big): " << big << endl;
18     cout << "Character (grade): " << grade << endl;
19     cout << "Boolean (isStudent): " << isStudent << endl;
20
21     return 0;
22 }
```

// Type Modifiers and Type Casting

Once you understand data types, the next step is learning how to **adjust and control them**. Sometimes you need to convert one type into another, and sometimes you need to expand or restrict the range of a data type. That's where **type casting** and **type modifiers** come in.

Type Casting in C++

What is Type Casting?

Type casting means **converting one data type into another**. In real life, imagine you have a bill amount ₹99.75, but you only want to store ₹99 as a whole number. You are converting a decimal into an integer. This is exactly what type casting does in C++. It changes the **type of a value** so it can be used differently.

Why Do We Need Type Casting?

Type casting is used:

- To perform accurate calculations
- To avoid unintended results
- To match data types in expressions

Example:

```
int a = 10, b = 3;  
float result = a / b;
```

This gives 3 (wrong for decimal division).

Correct way:

```
float result = (float)a / b;
```

Now the result is 3.33.

Types of Type Casting:

1. Implicit Type Casting (Automatic)

Done automatically by the compiler.

```
int a = 10;  
float b = a; // int → float
```

2. Explicit Type Casting (Manual)

Done by the programmer.

```
float x = 5.7;  
int y = (int)x; // float → int
```

Type Modifiers in C++

What are Type Modifiers?

Type modifiers are used to **modify the size or range of basic data types**.

Think of them like choosing different container sizes:

- Small values → short
- Large values → long
- Only positive values → unsigned

Why Do We Need Type Modifiers?

Because different data needs different storage:

- Marks → small values
- Population → large values
- Count → always positive

Modifiers help:

- Save memory
- Increase range
- Improve efficiency

Common Type Modifiers

- **short** → smaller range
- **long** → larger range
- **unsigned** → only positive values
- **signed** → both positive and negative values

Type Modifiers with Different Data Types

Modifiers are mainly used with **integer and character types**.

Examples:

```
short int a = 10;  
long int population = 2000000000;  
unsigned int marks = 100;  
signed int temp = -50;  
  
unsigned char ch = 255;
```

Lesson Program: Type Modifiers and Type Casting in C++**Source Code:** TypeModifying&Casting.cpp

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // ----- Type Casting -----
7
8      int a = 10, b = 3;
9
10     // Without type casting
11     float result1 = a / b;
12     cout << "Without Type Casting: " << result1 << endl;
13
14     // With type casting
15     float result2 = (float)a / b;
16     cout << "With Type Casting: " << result2 << endl;
17
18     // Explicit type casting
19     float x = 5.7;
20     int y = (int)x;
21     cout << "Float to Int: " << y << endl;
22
23     // ----- Type Modifiers -----
24
25     short int smallNum = 100;
26     long int bigNum = 2000000000;
27
28     unsigned int positiveNum = 40000;
29     signed int negativeNum = -200;
30
31     unsigned char ch = 250;
32
33     // Displaying modified types
34     cout << "Short int: " << smallNum << endl;
35     cout << "Long int: " << bigNum << endl;
36     cout << "Unsigned int: " << positiveNum << endl;
37     cout << "Signed int: " << negativeNum << endl;
38     cout << "Unsigned char: " << (int)ch << endl;
39
40     return 0;
41 }

```

Operators in C++

In C++, operators are symbols that tell the computer to perform operations on data. Whether it is adding two numbers, comparing values, or assigning a result, operators are what make programs *do work* instead of just holding values. In simple terms: **Operators are used to perform operations on variables and constants.** For example: `int sum = a + b;` Here, `+` is an operator that adds two values.

Types of Operators in C++

C++ provides different types of operators based on the kind of operation they perform.

1. Arithmetic Operators

- Perform basic mathematical operations.
- Operators: `+` (addition), `-` (subtraction), `*` (multiplication), `/` (division), `%` (modulus).
- Example (Math use):
 - `10 + 5 = 15`
 - `10 - 5 = 5`
 - `10 * 5 = 50`
 - `10 / 5 = 2`
 - `10 % 3 = 1`

2. Relational Operators

- Compare two values and return true (1) or false (0).
- Operators: `<`, `>`, `<=`, `>=`, `==`, `!=`.
- Example (Math use):
 - `10 > 5 → true`
 - `7 == 3 → false`

3. Logical Operators

- Used to combine relational expressions.
- Operators: `&&` (logical AND), `||` (logical OR), `!` (logical NOT).
- Example (Math use):
 - `(10 > 5) && (5 < 20) → true`
 - `(10 < 5) || (5 < 20) → true`
 - `!(10 > 5) → false`

4. Bitwise Operators

- Perform operations on binary representations of integers.
- Operators: `&` (AND), `|` (OR), `^` (XOR), `~` (NOT), `<<` (left shift), `>>` (right shift).
- Example (Binary use):
 - `5 & 3 → 1 (0101 & 0011 = 0001)`
 - `5 | 3 → 7 (0101 | 0011 = 0111)`

5. Assignment Operators

- Assign values to variables.
- Operators: `=`, `+=`, `-=`, `*=`, `/=`, `%=`.

- Example (Math use):
 - $a = 10$
 - $a += 5 \rightarrow a = a + 5$

6. Unary Operators

- Operate on a single operand.
- Operators: $++$ (increment), $--$ (decrement), $!$ (logical NOT), $\sim$ (bitwise NOT).
- Example (Math use):
 - $a = 5, ++a \rightarrow 6$
 - $a = 5, --a \rightarrow 4$

7. Ternary (Conditional) Operator

- Shorthand for if-else condition.
- Operator: $?:$
- Example (Math use):
 - $(10 > 5) ? 1 : 0 \rightarrow 1$

8. Miscellaneous Operators

- `sizeof()` $\rightarrow$ gives the size of a data type.
- `,` (comma) $\rightarrow$ separate expressions.
- `&` $\rightarrow$ address of variable.
- `*` $\rightarrow$ pointer to variable.

Operator Precedence and Associativity:

When an expression contains multiple operators, C++ follows certain rules to decide which operation to perform first.

Precedence	Operator	Description	Associativity
1	<code>()</code> <code>[]</code> <code>.</code> <code>-></code> <code>++</code> <code>--</code>	Parentheses or function call Brackets or array subscript Dot or Member selection operator Arrow operator Postfix increment/decrement	left to right
2	<code>++</code> <code>--</code> <code>+</code> <code>-</code> <code>!</code> <code>~</code> (type) <code>*</code> <code>&</code> <code>sizeof</code>	Prefix increment/decrement Unary plus and minus NOT operator and bitwise complement Type case Indirection or dereference operator Address of operator Determine size in bytes	right to left
3	<code>*</code> <code>/</code> <code>%</code>	Multiplication, division, and modulus	left to right
4	<code>+</code> <code>-</code>	Addition and subtraction	left to right
5	<code><<</code> <code>>></code>	Bitwise left shift and right shift	left to right
6	<code><</code> <code><=</code> <code>></code> <code>>=</code>	Relational less than and less than or equal to Relational greater than and greater than or equal to	left to right
7	<code>==</code> <code>!=</code>	Relational equal to and not equal to	left to right
8	<code>&</code>	Bitwise AND	left to right
9	<code>^</code>	Bitwise exclusive OR	left to right
10	<code> </code>	Bitwise inclusive OR	left to right
11	<code>&&</code>	Logical AND	left to right
12	<code> </code>	Logical OR	left to right
13	<code>?:</code>	Ternary operator	right to left
14	<code>=</code> <code>+=</code> <code>-=</code> <code>*=</code> <code>/=</code> <code>%=</code> <code>&=</code> <code>^=</code> <code> =</code> <code><<=</code> <code>>>=</code>	Assignment operator Addition/ subtraction assignment Multiplication/ division assignment Modulus and bitwise assignment Bitwise exclusive and inclusive OR Bitwise left shift and right shift	right to left
15		Comma operator	left to right

Lesson Program: Operators in C++**Source Code: Operators.cpp**

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      int a = 10, b = 5;
7
8      // ----- Arithmetic Operators -----
9      cout << "Arithmetic Operators:\n";
10     cout << "a + b = " << a + b << endl;
11     cout << "a - b = " << a - b << endl;
12     cout << "a * b = " << a * b << endl;
13     cout << "a / b = " << a / b << endl;
14     cout << "a % b = " << a % b << endl;
15
16     // ----- Relational Operators -----
17     cout << "\nRelational Operators:\n";
18     cout << "a > b: " << (a > b) << endl;
19     cout << "a < b: " << (a < b) << endl;
20     cout << "a == b: " << (a == b) << endl;
21     cout << "a != b: " << (a != b) << endl;
22
23     // ----- Logical Operators -----
24     cout << "\nLogical Operators:\n";
25     cout << "(a > 5 && b < 10): " << (a > 5 && b < 10) << endl;
26     cout << "(a < 5 || b < 10): " << (a < 5 || b < 10) << endl;
27     cout << "! (a > b): " << !(a > b) << endl;
28
29     // ----- Assignment Operators -----
30     cout << "\nAssignment Operators:\n";
31     int x = 10;
32     x += 5;
33     cout << "x += 5: " << x << endl;
34
35     x -= 3;
36     cout << "x -= 3: " << x << endl;
37
38     x *= 2;
39     cout << "x *= 2: " << x << endl;
40
41     x /= 4;
42     cout << "x /= 4: " << x << endl;
43
44     // ----- Increment / Decrement -----
45     cout << "\nIncrement / Decrement:\n";
46     int y = 5;
47     cout << "y = " << y << endl;
48     cout << "y++ = " << y++ << endl; // post-increment
49     cout << "After y++: " << y << endl;
50
51     cout << "++y = " << ++y << endl; // pre-increment
52     cout << "After ++y: " << y << endl;
53
54     // ----- Bitwise Operators -----
55     cout << "\nBitwise Operators:\n";
56     cout << "a & b = " << (a & b) << endl;
57     cout << "a | b = " << (a | b) << endl;
58     cout << "a ^ b = " << (a ^ b) << endl;
59     cout << "~a = " << (~a) << endl;
60     cout << "a << 1 = " << (a << 1) << endl;
61     cout << "a >> 1 = " << (a >> 1) << endl;
62
63     // ----- Conditional (Ternary) Operator -----
64     cout << "\nTernary Operator:\n";
65     int max = (a > b) ? a : b;
66     cout << "Max of a and b: " << max << endl;
67
68     // ----- sizeof Operator -----
69     cout << "\nsizeof Operator:\n";
70     cout << "Size of int: " << sizeof(a) << " bytes" << endl;
71     cout << "Size of float: " << sizeof(float) << " bytes" << endl;
72
73     // ----- Comma Operator -----
74     cout << "\nComma Operator:\n";
75     int m;
76     m = (a = 3, b = 4, a + b);
77     cout << "Result using comma operator: " << m << endl;
78
79     // ----- Address-of Operator -----
80     cout << "\nAddress-of Operator:\n";
81     cout << "Address of a: " << &a << endl;
82
83     return 0;
84 }

```

Input and Output in C++

A program becomes truly useful when it can **communicate** take input from the user and display output back. In C++, this communication is handled using **Input/Output (I/O)**. Instead of working silently, a program can ask questions, accept values, process them, and show results. This interaction is made possible through the **iostream library**.

What is iostream in C++?

iostream stands for Input Output Stream. It is a standard library in C++ that provides tools to:

- Take input from the user
- Display output on the screen

When you write: `#include <iostream>`

You are telling the compiler: "I want to use input and output features in my program". Without including **iostream**, you cannot use **cin** and **cout**.

What is a Stream?

A stream is a flow of data.

Think of it like a pipe :

- Input stream → data flows into the program
- Output stream → data flows out of the program

C++ uses streams to handle all input and output operations.

cout (Output Stream):

cout stands for **console output**. It is used to display output on the screen. "Send this data to the screen"

Example: `cout << "Hello";`

Here:

- `<<` is called the **insertion operator**
- It sends data to the output stream

When to Use cout?

- To display messages
- To show results
- To print values of variables

cin (Input Stream):

cin stands for **console input**. It is used to take input from the user.

Example: `cin >> age;`

Here:

- `>>` is called the **extraction operator**
- It takes input from the user and stores it in a variable

“Take input and store it in this variable”

When to Use cin?

- When user input is required
- When values are not fixed
- When making programs interactive

Basic Input/Output Flow

1. Program asks for input (`cout`)
2. User enters data (`cin`)
3. Program processes data
4. Program displays result (`cout`)

Common Errors in Input/Output

Beginners often face small but important mistakes:

1. Missing `#include <iostream>`

Error: `cin` or `cout` not recognized

2. Missing `using namespace std;`

Error: Need to write `std::cin`, `std::cout`

3. Using wrong operator

`cin << age;` // ❌ Wrong

`cout >> age;` // ❌ Wrong

Correct: `cin >> age;` `cout << age;`

4. Input Type Mismatch

`int age;`

`cin >> age;` // user enters text → error

Input must match variable type.

Think of cin and cout as basic conversation tools. The following features are like adding tone, formatting, and flexibility to that conversation.

1. `getline()` Taking Full Line Input

By default, cin reads input only **until the first space**. This becomes a problem when you want to read full names or sentences. It reads the entire line including spaces

Example: `string name; cin >> name;` // Input: John Doe → stores only "John"

To solve this, C++ provides `getline()`: `getline(cin, name);`

When to Use `getline()`?

- Full names
- Sentences
- Multi-word input

2. `endl` vs `\n`

Both are used to move to a new line, but they behave slightly differently.

- `\n` → just moves to next line, `cout << "Hello\n";`
- `endl` → moves to next line and flushes the output buffer, `cout << "World" << endl;`

For beginners, both are fine, but `\n` is slightly faster.

3. `iomanip` Formatting Output

C++ provides a library called `<iomanip>` for formatting output. Include it like:

`#include <iomanip>`

Common Formatting Functions:

1. `setw()` → Set Width

`cout << setw(10) << 100;`

Prints value with spacing

2. `setprecision()` → Decimal Precision

`cout << setprecision(3) << 3.14159;`

Controls number of digits

3. `fixed` → Fixed Decimal Format

`cout << fixed << setprecision(2) << 3.14159;`

Output: 3.14

4. `cerr` → Error Output

`cerr` is used to display error messages.

`cerr << "Error: Invalid input";`

It is unbuffered and prints immediately

5. `clog` → Logging Output

`clog` is used for logging messages.

`clog << "Log: Program started";`

Used for debugging or tracking program flow

Summary of Common I/O Tools

Tool	Purpose
cin	Input
cout	Output
getline()	Full line input
endl / \n	New line
setw()	Formatting width
setprecision()	Decimal control
cerr	Error messages
clog	Logging

Lesson Program: Input & Output in C++**Source Code: InputOutput.cpp**

```

1  #include <iostream>
2  #include <iomanip>
3  using namespace std;
4
5  int main() {
6
7      // ----- Basic Input using cin -----
8      string name;
9      int age;
10     float marks;
11
12     cout << "Enter your name (single word): ";
13     cin >> name;
14
15     cout << "Enter your age: ";
16     cin >> age;
17
18     cout << "Enter your marks: ";
19     cin >> marks;
20
21     // ----- Displaying Output -----
22     cout << "\n--- Student Details ---\n";
23     cout << "Name: " << name << endl;
24     cout << "Age: " << age << endl;
25     cout << "Marks: " << marks << endl;
26
27     // ----- Calculation -----
28     int nextAge = age + 1;
29     cout << "Next year, you will be: " << nextAge << endl;
30
31     // ----- getline() Example -----
32     string fullName;
33
34     cin.ignore(); // clear buffer before getline
35
36     cout << "\nEnter your full name: ";
37     getline(cin, fullName);
38
39     cout << "Full Name: " << fullName << endl;
40
41     // ----- Formatting Output -----
42     float value = 3.14159;
43
44     cout << "\nWithout formatting: " << value << endl;
45
46     cout << "Using setprecision(3): "
47         << setprecision(3) << value << endl;
48
49     cout << "Using fixed and setprecision(2): "
50         << fixed << setprecision(2) << value << endl;
51
52     // ----- setw() Example -----
53     cout << "\nUsing setw():\n";
54     cout << setw(10) << 100 << endl;
55     cout << setw(10) << 200 << endl;
56
57     // ----- cerr Example -----
58     cerr << "\nError: This is a sample error message\n";
59
60     // ----- clog Example -----
61     clog << "Log: Program executed successfully\n";
62
63     return 0;
64 }

```

// Other Input & Output Methods in C++

So far, we have used `cin` and `cout` from the `<iostream>` library for input and output. While they are the most common, C++ also provides **other ways to take input and display output**, especially through functions inherited from C and some additional utilities.

These methods are useful when:

- You need simpler or faster input/output
- You want character-based or formatted input
- You are working with legacy code or competitive programming

1. `scanf()` and `printf()` (from C)

C++ also supports C-style input/output using the `<stdio>` library. They are formatted I/O functions, meaning you must specify the type of data.

Include it like: `#include <stdio>`

What are `scanf()` and `printf()`?

- `scanf()` → used for input
- `printf()` → used for output

Example

```
int age;
scanf("%d", &age);
printf("Age = %d", age);
```

When to Use?

- Faster input/output (used in competitive programming)
- When working with C-style code

2. `getchar()` and `putchar()`

These are used for **single character input and output**.

What are they?

- `getchar()` → reads a single character
- `putchar()` → prints a single character

Example:

```
char ch;
ch = getchar();
putchar(ch);
```

When to Use?

- Reading character-by-character input
- Simple input/output tasks

3. gets() and puts() (Basic Idea):

These are used for string input/output (from C-style).

- gets() → reads a string
- puts() → prints a string

Note: gets() is **unsafe and not recommended** in modern C++.

Lesson Program: Input & Output other Methods in C++

Source Code: OtherI&O.cpp

```

1  #include <iostream>
2  #include <cstdio>
3  using namespace std;
4
5  int main() {
6
7      // ----- scanf and printf -----
8      int age;
9      float marks;
10
11     printf("Enter age: ");
12     scanf("%d", &age);
13
14     printf("Enter marks: ");
15     scanf("%f", &marks);
16
17     printf("Age = %d\n", age);
18     printf("Marks = %.2f\n", marks);
19
20     // ----- getchar and putchar -----
21     char ch;
22
23     printf("\nEnter a character: ");
24     getchar(); // to consume leftover newline
25     ch = getchar();
26
27     printf("You entered: ");
28     putchar(ch);
29     printf("\n");
30
31     // ----- puts -----
32     puts("\nThis is a string using puts");
33
34     return 0;
35 }
```

Program Flow in C++

When a C++ program runs, the computer follows a specific order to execute instructions. This order is called the **program flow** or **control flow** of the program.

By default, C++ executes statements **one after another from top to bottom**. The computer starts from the first executable statement and continues step by step until the program ends.

Here, the execution happens in a straight line:

1. First statement executes
2. Then second
3. Then third

This is called **sequential execution**.

Why Is Program Flow Important?

Real programs are rarely simple straight-line instructions. Sometimes a program needs to: make decisions, repeat actions, skip certain instructions, stop execution early, jump to another part of the program

Imagine a login system:

- If password is correct → allow access
- Otherwise → show error

Or think about a game:

- Repeat actions until the player loses
- Increase score continuously

Programs need flexibility. They cannot always move only in one direction like a train on a single track. This is where control flow statements become important.

What Are Control Flow Statements?

Control flow statements are special statements that change the normal execution flow of a program. They allow the program to:

- Choose between different paths
- Repeat blocks of code
- Interrupt execution
- Transfer control to another statement

Without control flow statements, programs would only execute instructions in fixed order and would never be able to make decisions or perform repetition.

Types of Program Flow in C++

Program execution in C++ mainly follows three kinds of flow:

1. Sequential Flow

This is the default flow. Statements execute one after another in the order they are written. Each line executes sequentially.

2. Selection (Decision) Flow

Sometimes the program needs to decide which path to follow. Example: If marks are above 40 → pass, Else → fail

This is done using:

- if
- if-else
- switch

3. Repetition (Looping) Flow

Sometimes a task must repeat multiple times. Example: Print numbers from 1 to 10, Keep asking for password until correct

This is done using:

- for
- while
- do-while

Flow Disruption in Programs

Control flow statements disrupt the normal top-to-bottom execution.

For example:

- Loops send execution backward repeatedly 
- Conditions choose one path among many 
- Jump statements like break or continue interrupt flow instantly ⚡

Because of this, program execution becomes dynamic instead of fixed.

Understanding Program Flow Visually

You can imagine program flow like water flowing through pipes 

- Normal execution → water flows straight
- Conditions → water chooses a branch
- Loops → water circulates repeatedly
- Jump statements → water suddenly changes direction

The programmer controls this flow using control statements.

Conditional Statements in C++

Conditional statements (also called decision-making statements) are used in C++ to make programs choose different paths of execution depending on whether a condition is true or false.

In real life, decisions happen constantly:

- If it rains → take an umbrella
- If marks are above 40 → pass
- If password is correct → allow login

Programs work the same way. They evaluate conditions and decide what should happen next. Conditional statements make programs more dynamic, intelligent, and responsive to different situations.

Types of Conditional Statements in C++

1. if Statement

The if statement is used to execute a block of code only when a condition is true.

Syntax:

```
if (condition) {  
    // code to execute if condition is true  
}
```

Example:

```
if (age >= 18) {  
    cout << "You can vote";  
}
```

2. if-else Statement

The if-else statement provides two-way decision making.

- If the condition is true → if block executes
- Otherwise → else block executes

Syntax:

```
if (condition) {  
    // code if condition is true  
} else {  
    // code if condition is false  
}
```

Example:

```
if (num % 2 == 0) {  
    cout << "Even";  
} else {  
    cout << "Odd";  
}
```

3. else-if Ladder

Used when multiple conditions need to be checked. The first true condition executes, and the remaining conditions are skipped.

Syntax:

```
if (condition1) {  
    // code  
} else if (condition2) {  
    // code  
} else {  
    // code if none are true  
}
```

Example:

```
if (marks >= 90) {  
    cout << "Grade A";  
} else if (marks >= 75) {  
    cout << "Grade B";  
} else {  
    cout << "Grade C or Fail";  
}
```

4. Nested if Statement

A nested if means an if statement inside another if statement. It is used for multiple levels of checking.

Syntax

```
if (condition1) {  
    if (condition2) {  
        // code if both are true  
    }  
}
```

Example

```
if (num > 0) {  
    if (num % 2 == 0) {  
        cout << "Positive Even Number";  
    }  
}
```

5. switch Statement

The switch statement is used when there are multiple fixed choices.

It works with:

- Integer values
- Character values
- Constant expressions

Instead of checking many conditions using else-if, switch directly jumps to the matching case.

Syntax

```
switch (expression) {  
    case value1:  
        // code  
        break;  
    case value2:  
        // code  
        break;  
    default:  
        // code if no case matches  
}
```

Example

```
switch (day) {  
    case 1:  
        cout << "Monday";  
        break;  
    case 2:  
        cout << "Tuesday";  
        break;  
    case 3:  
        cout << "Wednesday";  
        break;  
    default:  
        cout << "Invalid Day";  
}
```

Rules for switch Statement in C++

1. The switch expression must evaluate to:
 - int
 - char
 - enum
2. Case values must be unique constants.
3. break is optional but recommended.
 - With **break** → exits switch block
 - Without **break** → execution continues to next case (fall-through)
4. default executes if no case matches.

Closing Thought

Conditional statements give programs the ability to make decisions. Without them, programs would blindly execute every instruction in order. With them, programs can react, choose paths, and behave differently depending on situations. They are the beginning of real program logic.

Lesson Program: if & if-else statements in C++**Source Code:** ConditionalStatements1.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      int age, num;
7
8      // if statement example:
9      cout << "Enter your age: ";
10     cin >> age;
11
12     // Checking condition
13     if (age >= 18) {
14         cout << "You are eligible to vote.";
15     }
16
17     // if-else statement example:
18     cout << "Enter a number: ";
19     cin >> num;
20
21     // Checking even or odd
22     if (num % 2 == 0) {
23         cout << "The number is Even.";
24     } else {
25         cout << "The number is Odd.";
26     }
27
28     return 0;
29 }
```

Lesson Program: else-if ladder & nested if-else statements in C++**Source Code:** ConditionalStatements2.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      int marks, num;
7
8      // else-if ladder statement example:
9      cout << "Enter your marks: ";
10     cin >> marks;
11
12     // Checking grades
13     if (marks >= 90) {
14         cout << "Grade A";
15     } else if (marks >= 75) {
16         cout << "Grade B";
17     } else if (marks >= 50) {
18         cout << "Grade C";
19     } else {
20         cout << "Fail";
21     }
22
23     // Nested if-else statement example:
24     cout << "Enter a number: ";
25     cin >> num;
26
27     // Outer if
28     if (num > 0) {
29         // Inner if
30         if (num % 2 == 0) {
31             cout << "Positive Even Number";
32         } else {
33             cout << "Positive Odd Number";
34         }
35     } else {
36         cout << "Number is Zero or Negative";
37     }
38
39     return 0;
40 }
```

Lesson Program: switch statements in C++**Source Code:** ConditionalStatements3.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      int day;
7
8      cout << "Enter day number (1-7): ";
9      cin >> day;
10
11     // Switch statement example:
12     switch (day) {
13         case 1:
14             cout << "Monday";
15             break;
16         case 2:
17             cout << "Tuesday";
18             break;
19         case 3:
20             cout << "Wednesday";
21             break;
22         case 4:
23             cout << "Thursday";
24             break;
25         case 5:
26             cout << "Friday";
27             break;
28         case 6:
29             cout << "Saturday";
30             break;
31         case 7:
32             cout << "Sunday";
33             break;
34         default:
35             cout << "Invalid Day";
36     }
37
38     return 0;
39 }
```

Looping Statements in C++

Sometimes a program needs to execute the same block of code multiple times. Writing the same statements again and again would make programs lengthy and inefficient. To solve this, C++ provides **looping statements**.

Loops allow a program to repeat instructions until a condition becomes false. In real life, repetition happens everywhere:

- Printing numbers from 1 to 100
- Sending reminders every day
- Repeating game actions until the player loses

Loops make programs faster, cleaner, and more dynamic.

Types of Loops in C++

1. **for** Loop

The for loop is used when the number of repetitions is known beforehand. It combines: Initialization, Condition, Update all in a single line.

Syntax:

```
for (initialization; condition; update) {  
    // code to repeat  
}
```

Example:

```
for (int i = 1; i <= 5; i++) {  
    cout << i << endl;  
}
```

This prints numbers from 1 to 5.

2. **while** Loop

The while loop executes as long as the condition remains true.

It is mainly used when the number of repetitions is not fixed.

Syntax

```
while (condition) {  
    // code to repeat  
}
```

Example

```
int i = 1;  
while (i <= 5) {  
    cout << i << endl;  
    i++;  
}
```

3. do-while Loop

The do-while loop is similar to while, but with one important difference: The loop body executes at least once, even if the condition is false.

This happens because the condition is checked **after** execution.

Syntax:

```
do {  
    // code to repeat  
} while (condition);
```

Example:

```
int i = 1;  
do {  
    cout << i << endl;  
    i++;  
} while (i <= 5);
```

4. Nested Loops

A loop inside another loop is called a nested loop.

Nested loops are commonly used for:

- Patterns
- Tables
- Matrix operations

Syntax:

```
for (...) {  
    for (...) {  
        // inner loop code  
    }  
}
```

Example:

```
for (int i = 1; i <= 3; i++) {  
    for (int j = 1; j <= 2; j++) {  
        cout << "* ";  
    }  
    cout << endl;  
}
```

Output:

```
* *  
* *  
* *
```

Common Rules & Errors in Loops

Rules:

1. Always make sure the condition will eventually become false, otherwise → infinite loop.
2. Any of the three parts in `for(initialization; condition; update)` can be omitted, but semicolons must remain.
 - Example: `for(;;)` → infinite loop.
3. Variables declared in initialization are local to the loop.

Common Errors

- Missing semicolons in for or do-while.


```
do {
    ...
} while (condition) // ❌ error, needs ;
```
- Wrong condition leading to infinite loop.
- Forgetting update statement, causing infinite loop.
- Placing a semicolon immediately after loop (empty loop body).


```
for (i=0; i<5; i++); // ❌ does nothing
```

Feature	for loop	while loop	do-while loop
Condition checked	Before	Before	After
Executed at least once?	❌ No	❌ No	✅ Yes
Use case	Known iterations	Unknown iterations	Must run once

Closing Thought

Loops are what give programs rhythm 🔄 Without loops, programs would repeat code endlessly by hand. With loops, a few lines of code can perform thousands of operations automatically. They are one of the most powerful tools in programming because they allow repetition, automation, and efficiency.

Lesson Program: for loop statements in C++

Source Code: LoopingStatements1.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // for loop example 1:
7      // Printing numbers from 1 to 5
8      cout << "Numbers from 1 to 5:\n";
9
10     for (int i = 1; i <= 5; i++) {
11         cout << i << endl;
12     }
13
14     // for loop example 2:
15     // Printing even numbers
16     cout << "\nEven numbers from 2 to 10:\n";
17
18     for (int i = 2; i <= 10; i += 2) {
19         cout << i << endl;
20     }
21
22     return 0;
23 }
```

Lesson Program: while loop statements in C++

Source Code: LoopingStatements2.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // while loop example 1:
7      // Printing numbers from 1 to 5
8      int i = 1;
9
10     cout << "Numbers from 1 to 5:\n";
11
12     while (i <= 5) {
13         cout << i << endl;
14         i++;
15     }
16
17     // while loop example 2:
18     // Countdown
19     int count = 5;
20
21     cout << "\nCountdown:\n";
22
23     while (count >= 1) {
24         cout << count << endl;
25         count--;
26     }
27
28     return 0;
29 }
```

Lesson Program: do-while loop statements in C++**Source Code:** LoopingStatements3.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // do-while loop example 1:
7      // Printing numbers from 1 to 5
8      int i = 1;
9
10     cout << "Numbers from 1 to 5:\n";
11
12     do {
13         cout << i << endl;
14         i++;
15     } while (i <= 5);
16
17     // do-while loop example 2:
18     // Printing multiplication table of 2
19     int num = 1;
20
21     cout << "\nTable of 2:\n";
22
23     do {
24         cout << "2 x " << num << " = " << 2 * num << endl;
25         num++;
26     } while (num <= 10);
27
28     return 0;
29 }
```

Lesson Program: nested loop statements in C++

Source Code: LoopingStatements4.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // nested loop example 1:
7      // Printing square pattern
8      cout << "Square Pattern:\n";
9
10     for (int i = 1; i <= 3; i++) {
11         for (int j = 1; j <= 3; j++) {
12             cout << "* ";
13         }
14         cout << endl;
15     }
16
17     // nested loop example 2:
18     // Multiplication table
19     cout << "\nMultiplication Table:\n";
20
21     for (int i = 1; i <= 3; i++) {
22         for (int j = 1; j <= 3; j++) {
23             cout << i * j << "\t";
24         }
25         cout << endl;
26     }
27
28     return 0;
29 }
```

Jump Statements in C++

Jump statements in C++ are used to alter the normal flow of program execution. They allow the control of the program to move immediately from one part of the program to another.

Normally, programs execute sequentially from top to bottom. But sometimes we need to:

- Stop a loop early
- Skip certain iterations
- Jump directly to another part of the program
- Exit a function immediately

Jump statements provide this control.

Types of Jump Statements in C++

1. break Statement

The break statement is used to immediately terminate a loop or switch statement. Once break executes: Control moves outside the loop or switch block.

Where is break Used?

- for loop
- while loop
- do-while loop
- switch statement

Syntax: `break;`

2. continue Statement

The continue statement skips the current iteration of a loop and moves directly to the next iteration. Unlike break, it does not terminate the loop completely.

Where is continue Used?

- for loop
- while loop
- do-while loop

Syntax: `continue;`

3. goto Statement

The `goto` statement transfers control directly to a labeled statement inside the same function. It works like a jump in execution flow.

Syntax:

```
goto label;  
...  
label:  
statement;
```

Basic Idea of goto:

`goto` can make execution jump forward or backward in a program. Although it exists in C++, excessive use of `goto` is discouraged because it can make programs difficult to read and debug. This is often called **spaghetti code** 🍝

Modern programming usually prefers loops and functions instead of `goto`.

4. return Statement

The `return` statement is used to terminate a function and optionally send a value back. In `main()`, `return 0`; usually indicates that the program ended successfully.

Once `return` executes: The function immediately stops execution.

Syntax: `return value;` or `return;`

Closing Thought

Jump statements are like emergency doors and shortcuts inside a program:

- `break` exits
- `continue` skips
- `goto` jumps
- `return` ends execution

They give precise control over how a program moves and behaves. The real skill is not just knowing them, but knowing when a jump improves clarity and when it creates confusion.

Lesson Program: Jump statements in C++**Source Code:** JumpStatements.cpp

```

1  | #include <iostream>
2  | using namespace std;
3  |
4  | int main() {
5  |
6  |     // ----- break Statement -----
7  |     cout << "Break Statement:\n";
8  |
9  |     for (int i = 1; i <= 10; i++) {
10 |         if (i == 5) {
11 |             break;
12 |         }
13 |         cout << i << " ";
14 |     }
15 |
16 |     // ----- continue Statement -----
17 |     cout << "\n\nContinue Statement:\n";
18 |
19 |     for (int i = 1; i <= 5; i++) {
20 |         if (i == 3) {
21 |             continue;
22 |         }
23 |         cout << i << " ";
24 |     }
25 |
26 |     // ----- goto Statement -----
27 |     cout << "\n\nGoto Statement:\n";
28 |
29 |     goto message;
30 |
31 |     cout << "This line will be skipped\n";
32 |     cout << "This line will be skipped\n";
33 |     cout << "This line will be skipped\n";
34 |
35 | message:
36 |     cout << "Control jumped using goto";
37 |
38 |     // ----- return Statement -----
39 |     cout << "\n\nReturn Statement:\n";
40 |
41 |     int num = -5;
42 |
43 |     if (num < 0) {
44 |         cout << "Negative number detected. Program ending...\n";
45 |         return 0;
46 |     }
47 |
48 |     cout << "This line will not execute";
49 |
50 |     return 0;
51 | }

```

Arrays in C++

Until now, we have used variables to store single values.

Example: `int age = 21;`

But what if a program needs to store:

- Marks of 100 students
- Temperatures of 30 days
- Scores of multiple players

Creating separate variables for each value would become difficult and messy. This is where **arrays** become useful.

What is an Array?

An array is a collection of multiple values of the **same data type** stored under a single name. Instead of creating many variables, an array allows us to store related data together.

👉 An array is like a row of storage boxes 📦 📦 📦 Each box stores one value, and every box has a position number called an **index**.

Why Do We Need Arrays?

Arrays help:

- Store large amounts of related data
- Reduce number of variables
- Process multiple values easily using loops
- Organize data efficiently

Without arrays: `int mark1, mark2, mark3, mark4;`

With arrays: `int marks[4];`

Cleaner, simpler, and easier to manage.

How Arrays Work in Memory

Array elements are stored in **continuous memory locations**.

Each element has:

- Same data type
- Fixed position (index)

Example: `int marks[5];`

Memory representation:

```
marks[0]
marks[1]
marks[2]
marks[3]
marks[4]
```

Array indexing always starts from 0.

So:

- First element → index 0
- Second element → index 1

and so on.

Declaring an Array

Declaration means creating an array.

Syntax: `dataType arrayName[size];`

Example: `int marks[5];`

This creates an integer array capable of storing 5 values.

Initializing an Array

Initialization means assigning values to the array.

Syntax: `dataType arrayName[size] = {values};`

Example: `int marks[5] = {90, 85, 78, 92, 88};`

Here:

- `marks[0] = 90`
- `marks[1] = 85`
- and so on

Accessing Array Elements

Array elements are accessed using their index number.

Syntax: `arrayName[index]`

Example: `cout << marks[2];`

Output: 78

Because: `marks[2] = 78`

Modifying Array Elements

Values inside arrays can also be changed.

Example: `marks[1] = 95;`

Now the second element becomes 95.

Accessing All Array Elements Using Loops

Loops are commonly used with arrays because arrays usually contain multiple values.

Instead of writing:

```
cout << marks[0];  
cout << marks[1];  
cout << marks[2];
```

we use loops to access elements automatically.

Using for Loop:

```
for (int i = 0; i < 5; i++) {  
    cout << marks[i] << endl;  
}
```

Here:

- i acts as the index number
- Loop starts from 0
- Continues until last index

This prints all array elements one by one.

Using while Loop:

```
int i = 0;  
while (i < 5) {  
    cout << marks[i] << endl;  
    i++;  
}
```

Notes to Remember While Using Arrays

1. Array Index Starts from 0

For an array of size 5:

0 1 2 3 4

2. Array Size is Fixed

Once declared:

```
int arr[5];
```

Its size cannot be changed.

3. All Elements Must Be Same Type

```
int arr[5];
```

Can only store integers.

4. Accessing Invalid Index Causes Problems

```
arr[10]
```

This may cause unexpected behavior.

5. Loops are Commonly Used with Arrays

Loops make arrays powerful and practical.

Closing Thought:

Arrays are the first step toward handling large collections of data efficiently. Instead of thinking in terms of single values, arrays teach you to think in groups, sequences, and patterns. Once arrays enter programming, your programs begin to scale from handling one thing to handling many.

// Multidimensional Arrays

So far, we have worked with one-dimensional arrays, where data is stored in a single sequence. But in many real-world situations, data is organized in rows and columns instead of a simple line.

For example:

- Marks of students in different subjects
- Seating arrangement in a classroom
- Matrix calculations
- Game boards like chess or tic-tac-toe

To store and manage such structured data, C++ provides **multidimensional arrays**. A multidimensional array is simply an array that contains other arrays.

1D Arrays in C++

A one-dimensional array stores elements in a single row or sequence.

Example: `int numbers[5] = {10, 20, 30, 40, 50};`

Memory representation: `numbers[0] numbers[1] numbers[2] numbers[3] numbers[4]`

Only one index is needed to access elements.

Syntax of 1D Array: `dataType arrayName[size];`

Accessing Elements: `numbers[2]`

Output: 30

2D Arrays in C++

A two-dimensional array stores data in the form of **rows and columns**, similar to a table or matrix. You can think of it as: An array of arrays

Example: `int marks[2][3];`

This creates:

- 2 rows
- 3 columns

Memory representation:

`marks[0][0] marks[0][1] marks[0][2]`
`marks[1][0] marks[1][1] marks[1][2]`

Two indices are required:

- First index → row
- Second index → column

Declaring a 2D Array

Syntax: `dataType arrayName[rows][columns];`

Example: `int matrix[2][3];`

Initializing a 2D Array

Syntax:

```
dataType arrayName[rows][columns] = {  
    {values},  
    {values}  
};
```

Example:

```
int matrix[2][3] = {  
    {1, 2, 3},  
    {4, 5, 6}  
};
```

Accessing Elements in 2D Arrays

Syntax: `arrayName[row][column]`

Example: `cout << matrix[1][2];`

Output: 6

Because: `matrix[1][2] = 6`

Array Traversal

Traversal means visiting and processing every element of an array one by one. Loops are commonly used for traversal. Since 2D arrays contain rows and columns, nested loops are used.

Example:

```
for (int i = 0; i < 2; i++) {  
    for (int j = 0; j < 3; j++) {  
        cout << matrix[i][j] << " ";  
    }  
    cout << endl;  
}
```

Here:

- Outer loop → rows
- Inner loop → columns

Notes to Remember

1. Indexing Starts from 0

Both 1D and 2D arrays use zero-based indexing.

2. Arrays Store Same Data Type

A single array cannot mix integers, floats, characters, etc.

3. Nested Loops are Important for 2D Arrays

2D arrays are almost always used with nested loops.

4. Size Must Be Defined

The compiler needs array size during declaration.

Closing Thought

Arrays organize data. Multidimensional arrays organize relationships between data. Once arrays move from lines into rows and columns, programs begin handling information in a more structured and realistic way, much closer to how data exists in the real world.

Strings in C++

Until now, we have worked with numbers, characters, and arrays. But many real-world programs need to handle text.

For example:

- A student's name
- A city name
- A password
- A sentence entered by a user

Text data is represented using **strings**.

Strings are one of the most commonly used data types in programming because programs often need to store, process, compare, and display textual information.

What is a String?

A string is a sequence of characters stored together as a single unit.

Examples:

"Hello"

"Aditya"

"C++ Programming"

Each string consists of multiple characters arranged in order.

For example: "Hello"

contains: H e l l o

Why Do We Need Strings?

Suppose you want to store a person's name.

Without strings:

```
char c1 = 'A';
```

```
char c2 = 'd';
```

```
char c3 = 'i';
```

This quickly becomes impractical.

Instead: `string name = "Aditya";`

Strings allow us to store entire words, names, and sentences easily.

How Strings are Stored

A string is actually a collection of characters stored one after another in memory.

Example: "Hello"

Memory representation: H e l l o

Each character occupies its own memory location. This relationship between strings and characters is important because it explains why strings can be accessed character by character.

Strings in C++

C++ supports two major ways of working with strings:

1. Character Arrays (C-Style Strings)
2. String Class (std::string)

1. Character Arrays (C-Style Strings)

Before the C++ string class existed, strings were represented using arrays of characters.

Example: `char name[] = "Aditya";`

This creates a character array containing: A d i t y a \0

Null Character (\0): Every C-style string ends with a special character called the **null character**. '\0'

It tells the computer where the string ends. Without it, the program would not know where to stop reading characters.

Example: `char word[] = "Hello";`

Stored as: H e l l o \0

Advantages of Character Arrays

- Memory efficient
- Compatible with C programs
- Useful for low-level programming

Limitations of Character Arrays

- Fixed size
- Manual handling required
- String operations are more difficult

Example: `char name[20];`

The size must be specified in advance.

2. String Class (std::string)

Modern C++ provides the **string class**, which is easier and safer to use.

The string class belongs to the: <string> library.

Including String Library

```
#include <string>
```

Although many compilers include it automatically through <iostream>, it is good practice to include it explicitly.

Creating a String

```
string name = "Aditya";
```

or

```
string city;
```

Advantages of String Class

- Dynamic size
- Easy input/output
- Built-in functions
- Safer than character arrays
- Easier to manipulate text

Example:

```
string name = "Aditya";
```

```
cout << name;
```

Output: Aditya

Input and Output with Strings

Using cin:

```
string name;
```

```
cin >> name;
```

Input: Aditya

Limitation of cin:

cin stops reading at the first space.

Input: Aditya Varma

Stored: Aditya

Only the first word is stored.

Using getline()

To read complete lines: `string name;`
`getline(cin, name);`

Input: Aditya Varma

Stored: Aditya Varma

Accessing Characters in a String

Individual characters can be accessed using indexes.

Example:

```
string word = "Hello";  
cout << word[0];
```

Output: H

Length of a String

The string class can determine the number of characters stored.

```
string word = "Hello";  
cout << word.length();
```

Output: 5

Comparing Character Arrays and String Class

Character Array	String Class
Uses char[]	Uses string
Fixed size	Dynamic size
Manual operations	Built-in functions
More complex	Easier to use
Older C-style approach	Modern C++ approach

When to Use Which?

Use Character Arrays When:

- Learning how strings work internally
- Working with legacy C code
- Memory optimization is critical

Use String Class When:

- Writing modern C++ programs
- Handling user input
- Working with text data

Most modern C++ programs use the **string class**.

Notes to Remember

1. Strings are collections of characters.
2. Character arrays end with '\0'.
3. String class belongs to the <string> library.
4. cin reads only one word.
5. getline() reads entire lines.
6. String indexing starts from 0.

Example: string name = "CPLUSPLUS";

Indexes: 0 1 2 3 4 5 6 7 8
 C P L U S P L U S

Closing Thought

Strings allow programs to work with human language. While numbers help a program calculate, strings help it communicate. Understanding both C-style strings and the modern string class gives you a complete picture of how text is stored, processed, and manipulated in C++.